

Answers for the Report:

Q 1:

Answer (a) - Preprocessing:

- I removed rows with empty transcripts and converted all text to lowercase while removing punctuation to normalize the input.

Answer (b) - Feature computation:

- stop_words='english': to exclude common stop words that add little semantic value.
- max_df=0.8: to ignore words that appear in more than 80% of speeches as they are not informative.
- min_df=5: to keep words that occur in at least 5 speeches to filter out rare noise.
- max_features=1000: to select only the top 1000 most informative terms for dimensionality control.

Answer (c) - Number of features:

- I extracted 1000 features as specified by max_features, balancing richness and computational efficiency.

Q 2:

Answer (a) - Clustering algorithm:

I used the KMeans clustering algorithm because it is efficient for high-dimensional data like TF-IDF and provides centroids that can be interpreted through top features. It is also easy to control the number of clusters and compare results across different k values.

Answer (b) - Number of clusters:

- I selected 11 clusters, which gave the highest silhouette score during evaluation using a sample of 1000 instances.

Answer (c) - Evaluation:

- I used the silhouette score to measure how similar data points are within the same cluster versus other clusters. The best silhouette score achieved was 0.01968681406573233 for k = 11.

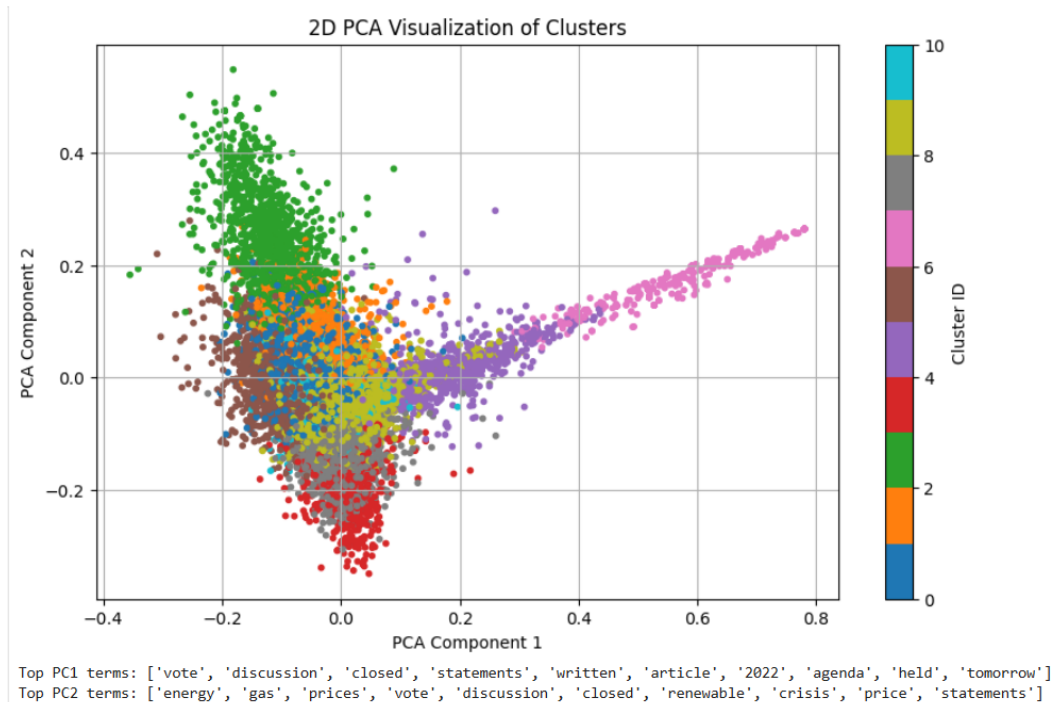
Answer (d) - Interpretation:

• Below are the top 10 most important words per cluster based on TF-IDF weights from the cluster centers:

- Cluster 0: european, member, commission, states, eu, union, digital, new, need, important
- Cluster 1: climate, food, need, farmers, european, production, sustainable, mr, change, europe
- Cluster 2: energy, gas, prices, european, need, renewable, price, crisis, electricity, nuclear
- Cluster 3: women, violence, rights, gender, girls, abortion, men, equality, sexual, afghanistan
- Cluster 4: agenda, point, session, request, parliament, voting, committee, meeting, 2022, discussion
- Cluster 5: ukraine, war, russia, putin, russian, european, ukrainian, people, europe, union
- Cluster 6: vote, closed, discussion, statements, written, article, 2022, held, debate, place
- Cluster 7: rights, law, human, rule, european, democracy, union, hungary, freedom, eu
- Cluster 8: european, mr, president, people, europe, think, parliament, union, want, colleagues
- Cluster 9: schengen, romania, bulgaria, croatia, european, area, border, accession, borders, union
- Cluster 10: health, young, children, youth, mental, people, cancer, european, care, education

Q3: PCA Visualization**Answer (a) - Your plot:**

• This figure which shows the debate transcripts in 2D PCA space, colored by their KMeans cluster assignments.



Answer (b) - Cluster separation:

- Some clusters show partial overlap, but a few groups are visually distinct in the PCA space.

Answer (c) - Interpretation:

- PCA-1 captures variance related to discussion topics like 'vote', 'discussion', 'statements', and 'article', likely summarizing procedural or parliamentary speech structure.
- PCA-2 highlights differences in content such as 'energy', 'gas', 'renewable', and 'crisis', representing economic and environmental subject matter.

Q4: Classification

Answer (a) - Preprocessing & feature transformations:

- I mapped the original win depth values into five numerical categories based on the provided rules. I also applied label encoding to the file and rank positions of all three pieces to convert them into numeric features suitable for classification.

Answer (b) - Model choice:

- I chose a Random Forest Classifier because it works well with encoded categorical data and performs robustly without heavy preprocessing. It also reduces overfitting by averaging over multiple decision trees.

Answer (c) - Evaluation setup:

- I split the data into 80% training and 20% testing and evaluated the model using accuracy, precision, recall, and F1-score with classification_report.

Answer (d) - Hyperparameters:

- I used GridSearchCV with 3-fold cross-validation to find the best combination of n_estimators and max_depth. The best-performing parameters were: n_estimators = 100, max_depth = None.

Answer (e) - Results:

- Below is the classification report showing the model's performance on the test set:

	precision	recall	f1-score	support
0	0.886	0.847	0.866	549.000
1	0.991	0.838	0.908	130.000
2	0.939	0.918	0.928	636.000
3	0.937	0.941	0.939	2072.000
4	0.942	0.963	0.952	2225.000
accuracy	0.935	0.935	0.935	0.935
macro avg	0.939	0.901	0.919	5612.000
weighted avg	0.935	0.935	0.935	5612.000